

Catalogue(a library - books)

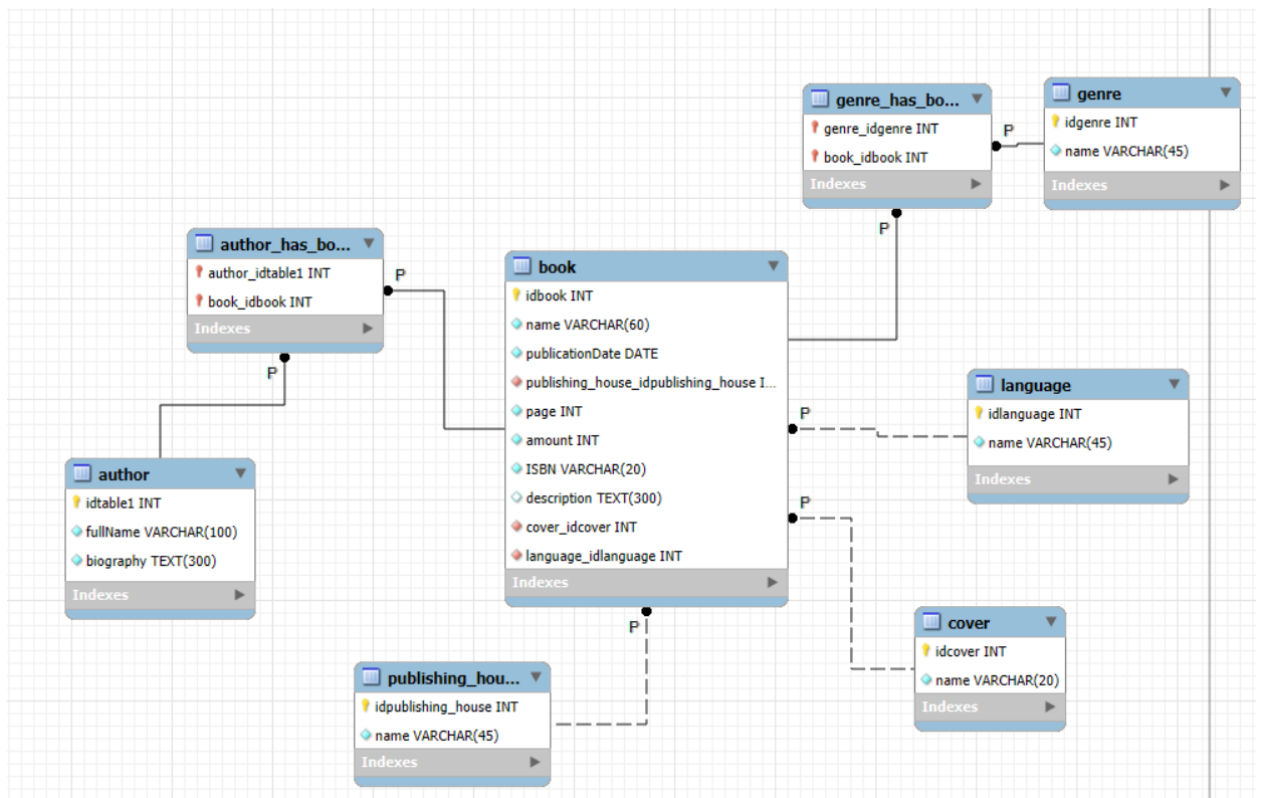
Requirements

Functional requirements:

1. The system must allow users to retrieve a paginated list of books;
2. The system must allow users to filter books by name, language, publishing house, cover type, author, genre, and page count;
3. The system must allow users to search for books by name or ISBN;;
4. The system must allow the admin to delete book, language etc
5. The system must allow users to retrieve detailed information about a single book, author, genre, or publisher by ID;
6. The system must allow the admin to add new books, authors, genres, publishing houses, languages, and cover types ;
7. The system must allow admins to delete books, authors genres etc. from the catalogue.

Non-Functional requirements:

1. **Security & Authorization.** Access to modification endpoints (POST, PUT, PATCH, DELETE) must be restricted to authenticated users with the ADMIN role;
2. **Error Handling & Reliability.** In the event of invalid user input, missing resources, or authorization failures, the system must return meaningful HTTP status codes;
3. **Performance & Caching.** Read operations (GET) for static and semi-static catalogue data must support HTTP caching headers;
4. **Pagination.** All list responses returning multiple resources must support pagination



book

- **idbook** — unique identifier of the book: INT NOT NULL
- **name** — title of the book: VARCHAR(60) NOT NULL
- **publicationDate** — date when the book was published: DATE NOT NULL
- **publishing_house_idpublishing_house** — foreign key referencing the publisher: INT NOT NULL
- **page** — total number of pages: INT NOT NULL
- **amount** — quantity of book copies available in stock: INT NOT NULL
- **ISBN** — International Standard Book Number: VARCHAR(20) NOT NULL
- **description** - description of the book: TEXT(300)
- **cover_idcover** — foreign key referencing the cover type: INT NOT NULL
- **language_idlanguage** — foreign key referencing the language of the book: INT NOT NULL

author

- **idauthor** — unique identifier of the author: INT NOT NULL
- **fullName** — author's full name: VARCHAR(100) NOT NULL
- **biography** — short biography of the author: TEXT(300) NOT NULL

genre

- **idgenre** — unique identifier of the genre: INT NOT NULL
- **genreName** — name of the genre (e.g., Fantasy, Sci-Fi): VARCHAR(45) NOT NULL

publishing_house

- **idpublishing_house** — unique identifier of the publishing house: INT NOT NULL
- **name** — name of the publisher: VARCHAR(45) NOT NULL

language

- **idlanguage** — unique identifier of the language: INT NOT NULL
- **name** — language name (e.g., English, Ukrainian): VARCHAR(45) NOT NULL

cover

- **idcover** — unique identifier of the cover type: INT NOT NULL
- **name** — type of binding/cover (e.g., Hardcover, Paperback): VARCHAR(20) NOT NULL

Operations:

GET - /books — Browse the book catalogue with support for search, filtering, sorting, and pagination

Query Params:

- **Filter:** language_id, publishing_house_id, cover_id, author_id, genre_id, min_pages, max_pages
- **Search:** search by name and ISBN
- **Order By :** sort (publicationDate:asc/desc, name:asc/desc)
- **Pagination:** page (from 1), limit (10)

- **Caching:** Cache-Control: public, max-age=300
- **Responses:** 200 OK, 400 Bad Request

GET - /books/{id} — Retrieve detailed information about a specific book by id.

- **Responses:** 200 OK, 404 Not Found

POST - /books — Add a new book to the catalogue (*Admin only*).

- **Responses:** 201 Created, 400 Bad Request, 401 Unauthorized, 403 Forbidden

PUT - /books/{id} — Completely replace all fields of a book by its id (*Admin only*).

- **Responses:** 200 OK, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found

PATCH - /books/{id} — Partially modify specific attributes of a book (e.g., only amount or description) (*Admin only*).

- **Responses:** 200 OK, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found

DELETE - /books/{id} — Remove a book from the system (*Admin only*).

- **Responses:** 204 No Content, 401 Unauthorized, 403 Forbidden, 404 Not Found

GET - /authors — Browse the list of authors.

- **Query Params:**
 - **Search:** fullName
 - **Order By:** sort (fullName:asc/desc)
 - **Pagination:** page, limit
- **Caching:** Cache-Control: public, max-age=86400
- **Responses:** 200 OK, 400 Bad Request

GET - /authors/{id} — Retrieve detailed information and biography of a specific author using id.

- **Responses:** 200 OK, 404 Not Found

POST - /authors — Create a new author entry (*Admin only*).

- **Responses:** 201 Created, 400 Bad Request, 401 Unauthorized, 403 Forbidden

GET - /genres — Browse the complete list of genres.

- **Query Params:**
 - **Search:** genreName
 - **Pagination:** page, limit
- **Caching:** Cache-Control: public, max-age=86400
- **Responses:** 200 OK

POST - /genres — Create a new genre entry (*Admin only*).

- **Responses:** 201 Created, 400 Bad Request, 401 Unauthorized, 403 Forbidden

GET - /publishing-houses — Browse the list of publishing houses.

- **Query Params:**
 - **Search:** name
 - **Pagination:** page, limit
- **Responses:** 200 OK

GET - /languages — List available book languages (e.g., English, Ukrainian).

- **Responses:** 200 OK

GET - /covers — List cover/binding types (e.g., Hardcover, Paperback).

- **Responses:** 200 OK